

Chapter 1: Introduction to Artificial Intelligence

4 hours | 7 marks

"Any sufficiently advanced technology is indistinguishable from magic."

— Arthur C. Clarke

AI is not magic. But once you understand it, you'll see *why* it looks like it.

The Hook — Why Does This Matter?

Before we define anything, let me ask you something:

How do you recognize your friend's face in a crowd?

You don't consciously say: *"Check eye distance... check nose shape... run facial geometry algorithm..."*

You just *know*. Instantly. Effortlessly.

Now ask yourself: **can a machine do that?**

That question — *can machines think, perceive, and reason like humans?* — is the entire field of Artificial Intelligence.

And the answer, it turns out, is: **yes, but differently.**

1.1 Defining Artificial Intelligence

The 4 Competing Definitions

There is no single definition everyone agrees on. Instead, there are four schools of thought (from Russell & Norvig's classic framework):

THINKING LIKE	ACTING LIKE
① Thinking Humanly (Cognitive Science)	② Thinking Rationally (Laws of Thought)
"Machines that model human thought processes"	"Machines that use logical reasoning"
③ Acting Humanly (The Turing Test)	④ Acting Rationally (Rational Agent)
"Machines that behave indistinguishably from humans"	"Machines that do the right thing to achieve their goals"

Modern AI (the field today) mostly uses definition ④ — the Rational Agent approach.

This is because it doesn't require machines to *imitate* humans — just to *achieve goals intelligently*.

The Definition We'll Use

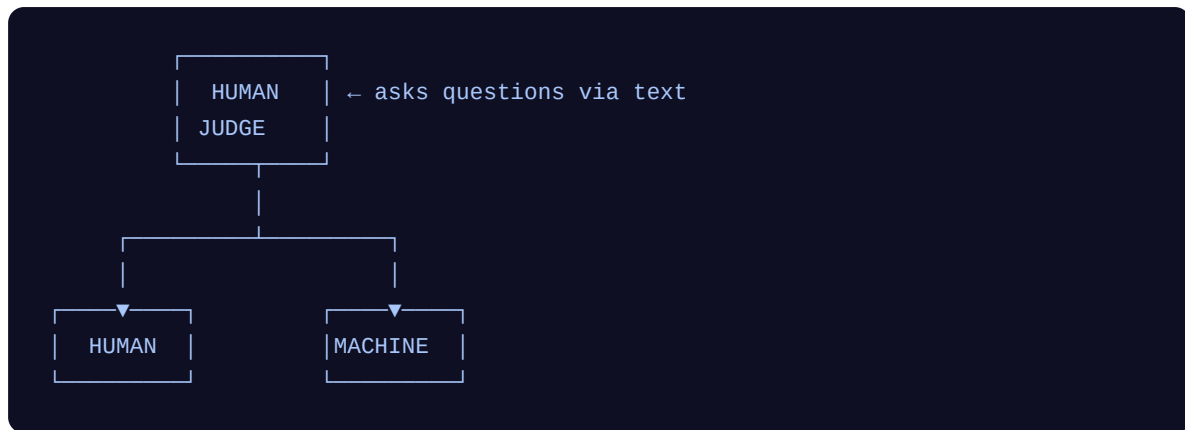
Artificial Intelligence is the science and engineering of making machines that can perceive their environment, reason about it, and take actions that maximize their chances of successfully achieving their goals.

1.2 The Turing Test — A Historical Touchstone

In 1950, mathematician **Alan Turing** asked: "*Can machines think?*"
He realized that's philosophically unanswerable. So he reframed it:

"Can a machine behave so intelligently that a human interrogator cannot tell it apart from another human?"

This became the **Turing Test** (he called it the "Imitation Game").



The judge asks both freely. If they can't tell who is the machine — the machine passes.

No machine has definitively passed the Turing Test (properly administered).

But here's the important philosophical point:

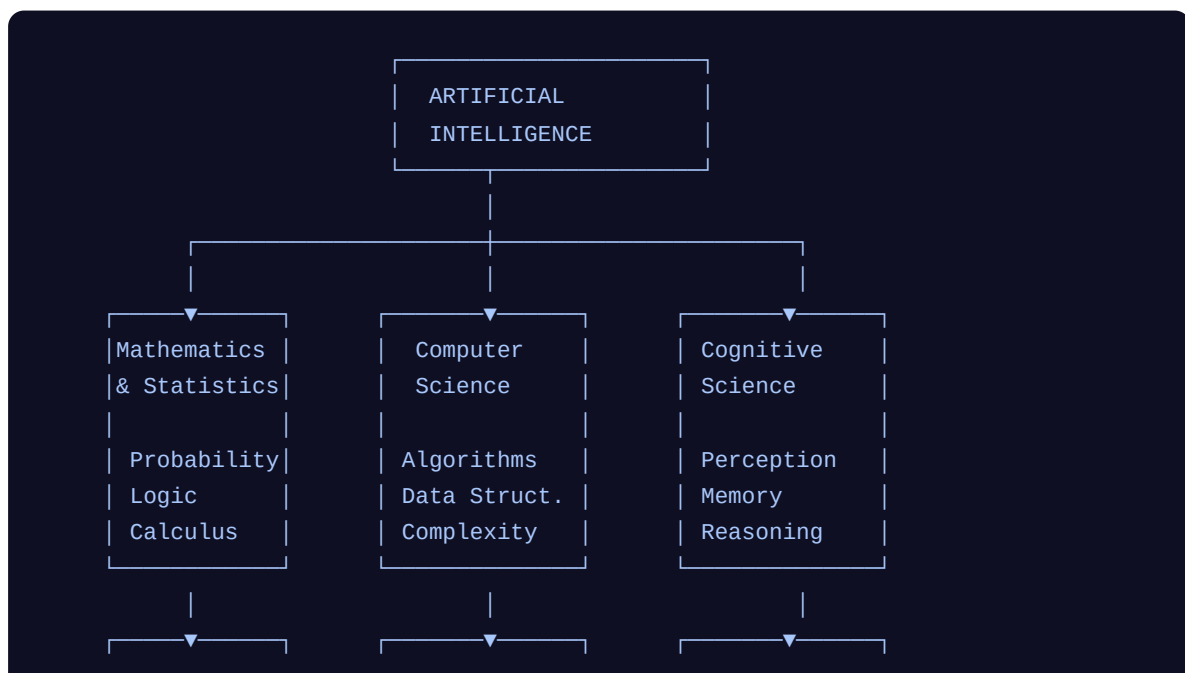
The Turing Test tests behavior — not understanding.

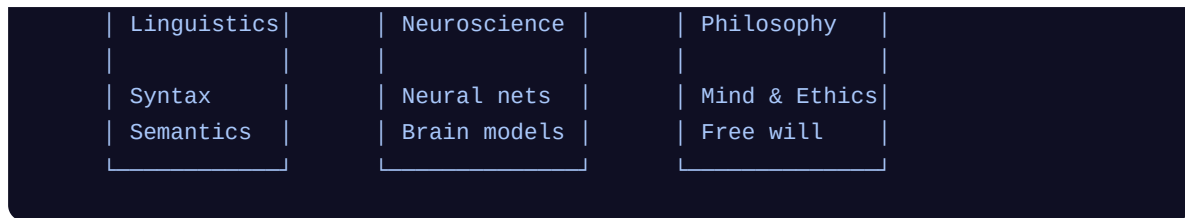
A machine could pass by being a very good mimic without truly "thinking."

This led to John Searle's famous **Chinese Room argument** — a machine can manipulate symbols perfectly without understanding their meaning. 🤖

1.3 AI and Related Fields — The Family Tree

AI does not exist in isolation. It is a *confluence* of many disciplines:





Key Sub-fields of AI

Sub-field	What it does	Example
Machine Learning	Systems learn from data without explicit programming	Spam filter
Natural Language Processing	Understanding and generating human language	ChatGPT
Computer Vision	"Seeing" and interpreting images/video	Face unlock
Robotics	Physical agents that act in the world	Warehouse robots
Expert Systems	Rule-based reasoning like a human expert	Medical diagnosis
Planning & Search	Finding optimal sequences of actions	GPS navigation

1.4 Brief History of AI

The Timeline (with context)

Let's not just list dates — let's understand *why* AI went through waves of excitement and despair.

- 1943 — McCulloch & Pitts
First mathematical model of a "neuron"
💡 Insight: The brain is computational!
- 1950 — Alan Turing
"Computing Machinery and Intelligence"
The Turing Test proposed
💡 Insight: Intelligence can be measured behaviorally
- 1956 — Dartmouth Conference

The word "Artificial Intelligence" coined
McCarthy, Minsky, Shannon, and others meet
💡 The field is officially born!

1956-1974 — GOLDEN AGE

Early programs solve algebra, geometry, play checkers
ELIZA (1966): First chatbot – shockingly convincing
💡 Optimism: "Human-level AI in 20 years!"

1974-1980 — FIRST AI WINTER ❄️

Computers weren't powerful enough
Problems were harder than expected
Funding cuts from government
💡 Lesson: Intelligence is harder than it looks

1980-1987 — EXPERT SYSTEMS BOOM

Rule-based systems for medical diagnosis, etc.
MYCIN diagnosed blood infections better than some doctors!
💡 Insight: Codify expert knowledge → AI

1987-1993 — SECOND AI WINTER ❄️

Expert systems too brittle, expensive to maintain
"AI" became a dirty word in business
💡 Lesson: Rules alone aren't enough

1993-2011 — QUIET REVOLUTION

Statistical ML emerges
Deep Blue beats Kasparov (1997) 🏆
Internet creates massive datasets
Support Vector Machines, Hidden Markov Models rise

2012 — THE DEEP LEARNING REVOLUTION 🚀

AlexNet wins ImageNet by a massive margin
Neural networks finally work – GPUs + Big Data
💡 This changed EVERYTHING

2016 — AlphaGo beats world Go champion
Go was considered "too complex" for machines

2017 — Transformer architecture invented
This eventually leads to GPT, BERT, ChatGPT

2022-Now — THE AGE OF FOUNDATION MODELS

ChatGPT, Gemini, Claude, DALL-E
AI enters everyday life

🎯 Key Lesson from History

AI progressed in waves because:

1. Initial optimism → hit fundamental limits
2. Better hardware + more data → new breakthroughs
3. The pattern repeats but each wave goes higher

This tells you something profound: **AI is not a single invention — it's a evolving science.**

1.5 Applications of AI — The Real World

Where AI Is Right Now (not sci-fi, actual reality)

Domain	Application	How AI Helps
Healthcare	Disease diagnosis from X-rays	Computer vision reads scans
Finance	Fraud detection	Anomaly detection in transactions
Language	Machine translation	NLP (Google Translate)
Gaming	AlphaGo, Chess engines	Search + ML
Autonomous Vehicles	Self-driving cars	Computer vision + planning
Agriculture	Crop disease detection	Image classification
Customer Service	Chatbots	NLP + retrieval
Science	Protein structure (AlphaFold)	Deep learning

The Practical Example: How Gmail filters spam

1. **Perception:** Gmail receives an email
2. **Feature extraction:** It notices words like "CLICK NOW", "FREE MONEY", strange sender
3. **Classification:** Learned model says "97% probability this is spam"
4. **Action:** Moves to spam folder

That's AI — **perceive** → **reason** → **act**.

1.6 Knowledge and Learning — The Two Pillars

This is philosophically the deepest part of Chapter 1.

What is Knowledge?

Knowledge is information that an agent can use to make decisions.

There are different kinds:

Type	Example	How Stored in AI
Declarative	"Kathmandu is the capital of Nepal"	Facts in a database
Procedural	"How to tie a shoelace"	Rules / programs
Heuristic	"If it looks like spam, it probably is"	Learned patterns
Meta-knowledge	"I know that I don't know X"	Uncertainty modeling

What is Learning?

Learning is the process by which an agent improves its performance based on experience.

Think of it this way:

WITHOUT LEARNING:

Robot encounters locked door → fails → stays failed

WITH LEARNING:

Robot encounters locked door → fails → updates model

→ next time, tries a key → succeeds

→ next time, goes straight to the key

Three fundamental types of learning (preview for Chapter 6):

1. **Supervised Learning** — "Learn from labeled examples" (like flashcards with answers)

2. **Unsupervised Learning** — "Find patterns in unlabeled data" (like grouping similar photos)
3. **Reinforcement Learning** — "Learn from trial and error + rewards" (like training a dog)



Math Intuition Corner

The Rationality Equation

A rational agent maximizes its **expected utility**:

$$\text{Expected Utility} = \sum P(\text{outcome} \mid \text{action}, \text{state}) \times U(\text{outcome})$$

Where:

- $P(\text{outcome} \mid \text{action}, \text{state})$ = probability of this outcome
- $U(\text{outcome})$ = how desirable (utility) this outcome is
- $\sum$ = sum over all possible outcomes

Intuition in plain English:

"Pick the action that, on average, gives you the best result, accounting for all possible ways things could go."

This is exactly what you do when deciding whether to carry an umbrella:

- You check the probability of rain (P)
- You weigh how bad getting wet is vs. how annoying carrying umbrella is (U)
- You pick the option with the best expected outcome

AI does exactly this, just faster and at scale.



Video Resources for This Chapter

Topic	Video	Why Watch
What is AI? History and Overview	The Turing Test — Crash Course AI	Clear overview, great animation

Topic	Video	Why Watch
The Turing Test in depth	The Turing Test explained — Tom Scott	Philosophical depth, highly engaging
History of AI	Brief History of AI — ColdFusion	Narrative storytelling of AI history
3Blue1Brown Neural Intro	But what is a neural network? — 3Blue1Brown	Builds the visual foundation you'll need

Worked Examples

Example 1: Is this AI?

A thermostat turns on heating when temperature drops below 20°C.

Is this AI?

✗ No. It follows a fixed rule with no perception beyond temperature, no learning, no reasoning. It's a simple control system.

Now upgrade it: a "smart" thermostat that:

- Learns your schedule
- Predicts when you'll be home
- Optimizes for energy savings + comfort
- Adapts to your feedback

✓ Yes! Now it perceives, reasons, learns, and acts toward a goal.

Example 2: Classify these as AI sub-fields

System	Sub-field
GPS route planning	Search / Planning
Siri understanding your voice	NLP + Speech Recognition

System	Sub-field
Doctor AI diagnosing chest X-ray	Computer Vision
Stock trading bot	ML + Planning
MYCIN (1970s medical system)	Expert Systems

Example 3: Applying the 4 Definitions

Scenario: A chess engine defeats a grandmaster.

Definition	Does Chess Engine Count?
Thinks humanly	✗ No — it doesn't reason like humans do
Thinks rationally	✓ Yes — uses logical rules of chess
Acts humanly	⚠ Partially — behavior looks intelligent
Acts rationally	✓ Yes — it picks optimal moves to win

⚡ Exam-Ready Summary

Must-Know Definitions

- **AI:** Science of making rational agents that perceive, reason, and act
- **Turing Test:** Behavioral test — can a machine fool a human judge?
- **Rational Agent:** Entity that takes actions to maximize expected utility
- **Knowledge:** Information usable for decision-making
- **Learning:** Improving performance through experience

Key Historical Milestones

Year	Event
1950	Turing Test proposed

Year	Event
1956	"AI" coined at Dartmouth
1997	Deep Blue beats Kasparov
2012	Deep Learning revolution (AlexNet)
2022	ChatGPT — AI goes mainstream

The 4 AI Definitions Table (exam favorite!)

Approach	Focus	Representative View
Think Humanly	Cognitive modeling	Brain simulation
Think Rationally	Laws of Thought	Logic
Act Humanly	Turing Test	Behavior
Act Rationally	Rational Agent	Modern AI

Types of AI Knowledge

- **Declarative** — facts ("what")
- **Procedural** — how to do things ("how")
- **Heuristic** — rules of thumb from experience

✓ Chapter 1 Checklist

Before moving to Chapter 2, make sure you can answer:

- ☐ Define AI in your own words using first principles
- ☐ Explain the Turing Test and its limitations (Chinese Room argument)
- ☐ Draw the 4-quadrant definition framework from memory
- ☐ List 5 applications of AI with the sub-field each belongs to
- ☐ Explain the difference between knowledge and learning
- ☐ Briefly describe the major phases of AI history and why "winters" happened

- ☐ Explain what a "rational agent" is using the expected utility concept
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